# Gene dosage architecture across complex traits and common diseases

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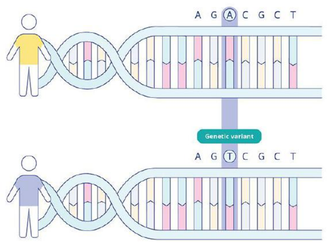
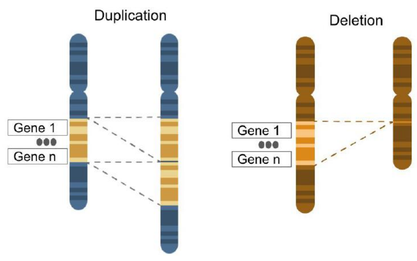
## Introduction



**Common Variants**

- Small effect sizes on complex traits
- Detected in >1% of population
- Well studied in research




**Rare Variants**
**Copy Number Variants (CNVs)**

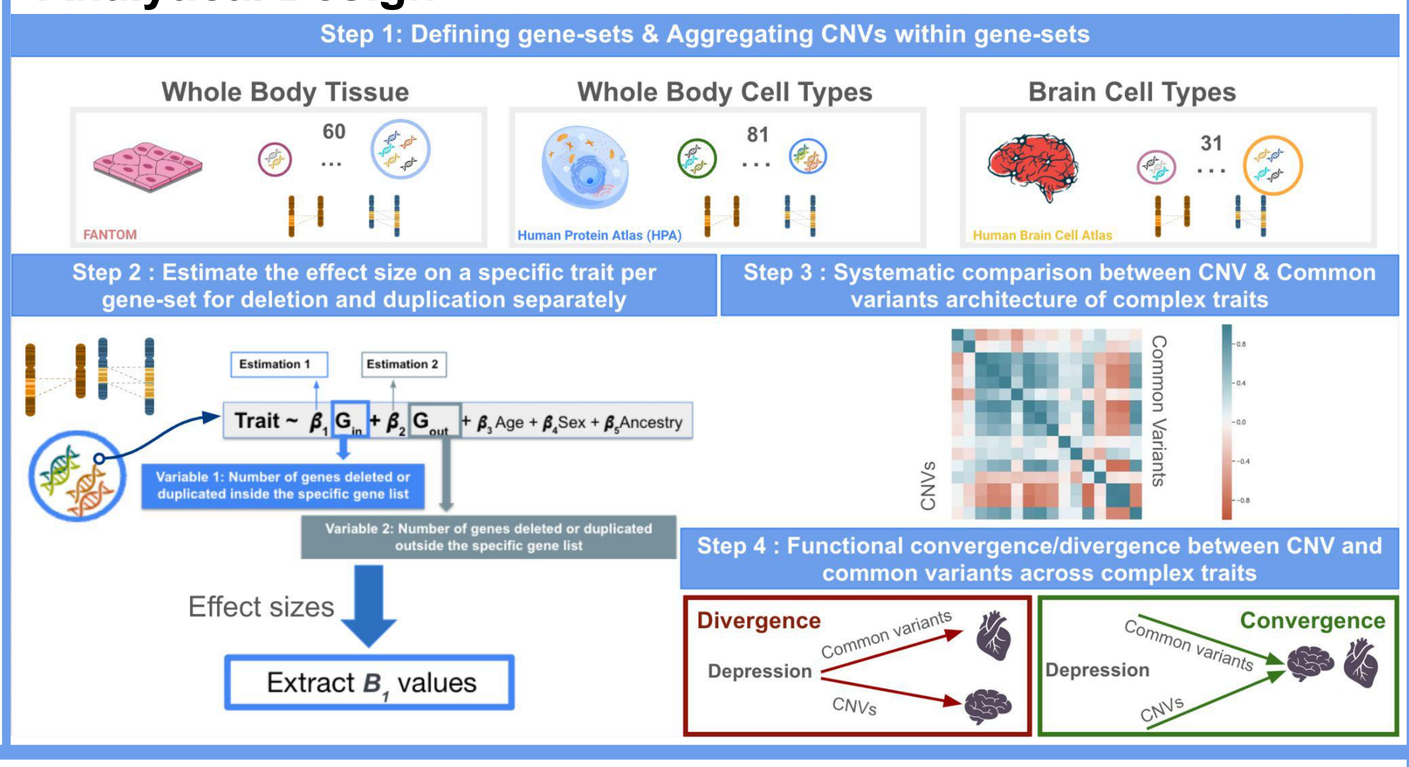
- Large effects on complex traits
- Detected in <1% of population
- Identified in clinics but not well studied in research.

- In human genetics, deciphering the functions of cells and tissues that mediate genetic associations with complex traits and diseases is a central question.

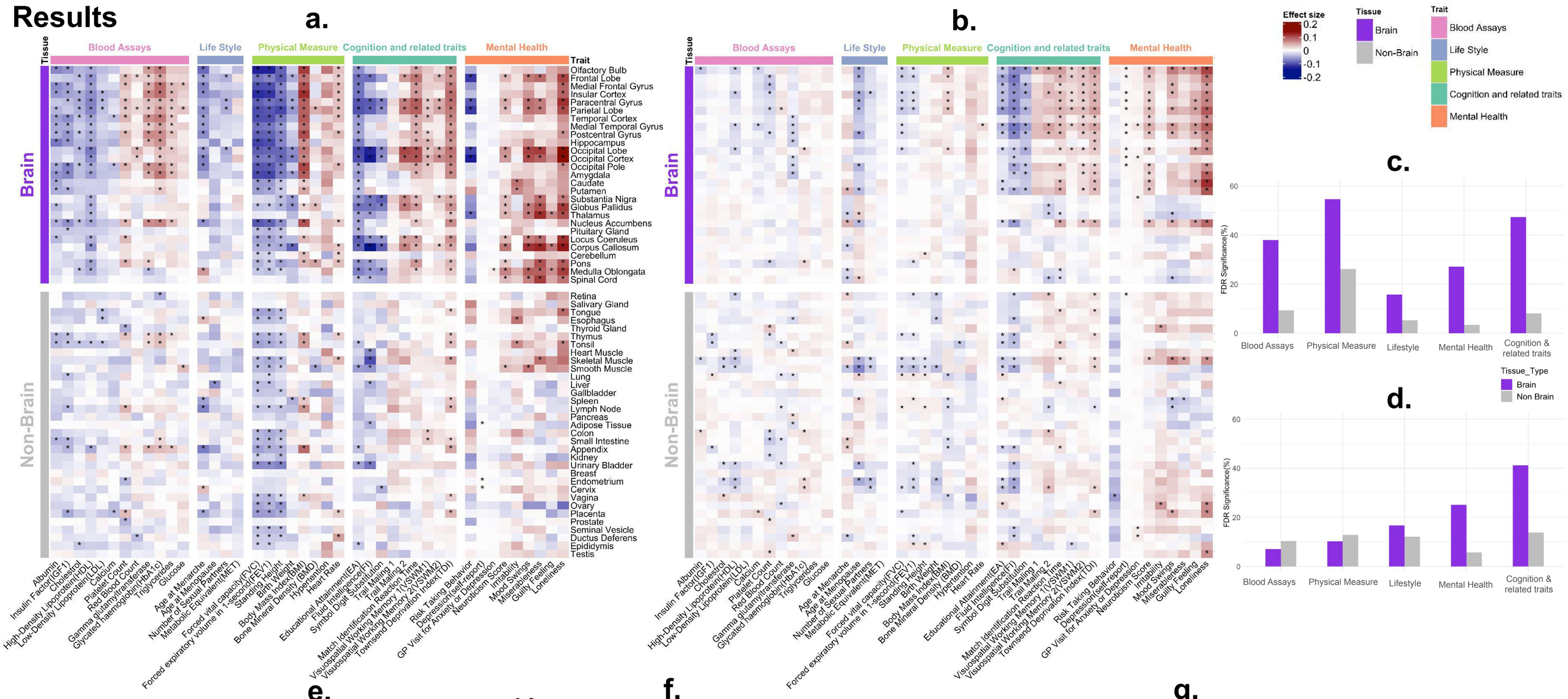
### Objectives

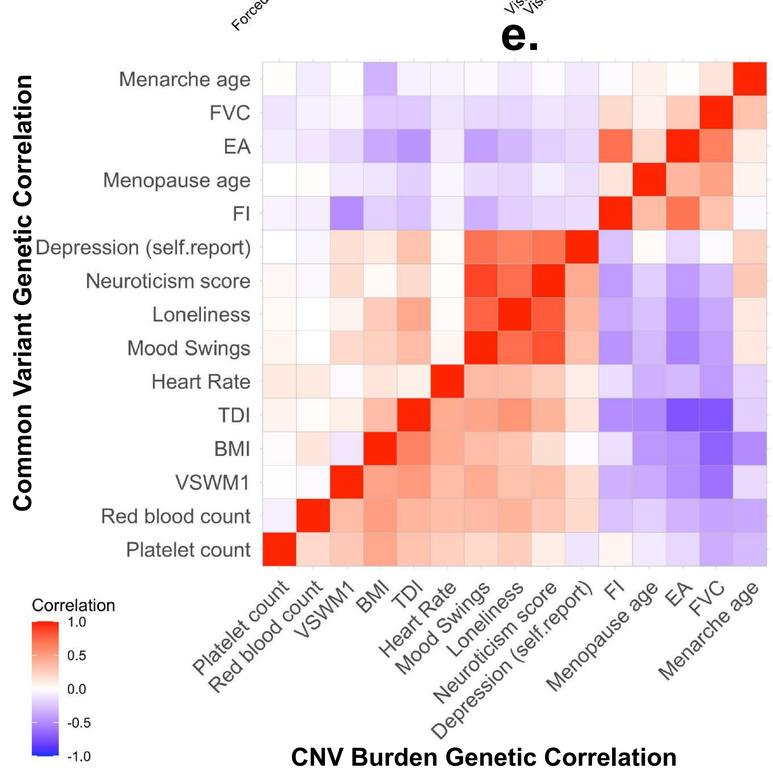
- Characterizing the CNV architecture of complex traits with mediation of functions
- Systematic comparison between CNV & Common variants architecture of complex traits

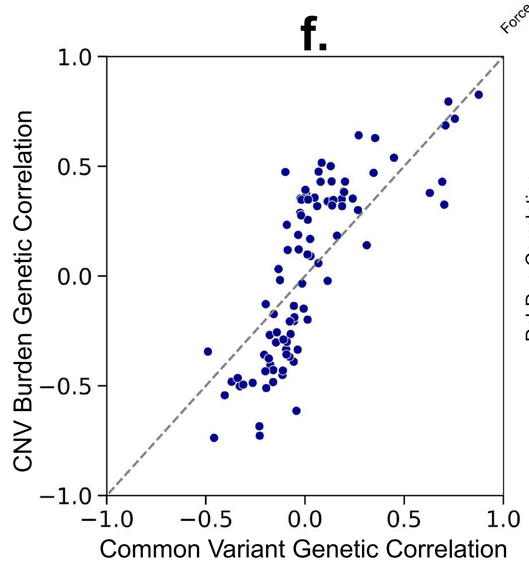
## Analytical Design




## Results



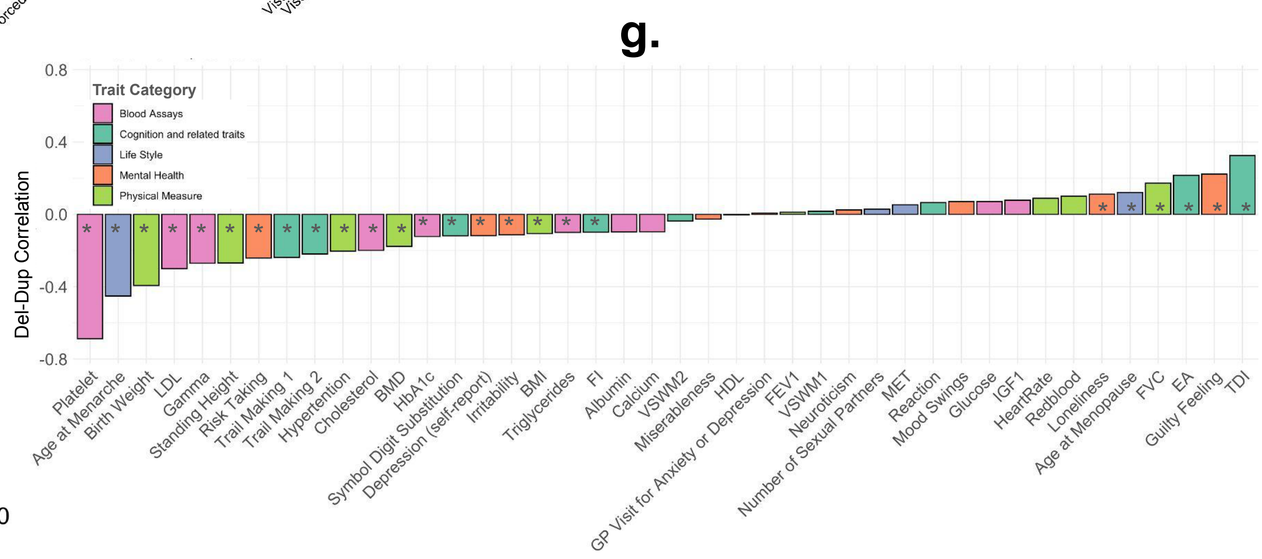




**Figure:** The panel **a** illustrates the effect sizes of 60 tissues on 42 traits for deletion. The traits and tissues are categorized and displayed along the x and y axes. The color intensity represents the effect size($B_1$) of the traits on the tissues. Black asterisks (*) indicate statistically significant associations after False Discovery Rate (FDR) correction. The same thing illustrated in panel **b** for duplication. Panel **c** and **d** shows the bar plots with the percentage of FDR significance within each trait and tissue categories, for deletion and duplication respectively. The color represents the tissue category. Panel **e** illustrates the heatmap of pairwise genetic correlation between traits using CNV in the lower triangle and common variant in the upper triangle. The color scale in the heatmap indicates the magnitude of the Spearman correlation between the traits. Panel **f** shows the scatterplot of the pairwise genetic correlations comparing common variants (x-axis) with CNVs (y-axis) (from panel e). Each point represents a pairwise correlation between traits, with the dashed line indicating perfect concordance (y = x). Panel **g** illustrates the Spearman correlation between Deletion(Del) and Duplication(Dup) effect size profiles(y-axis) across the traits(x-axis). Black asterisks (*) indicate statistically significant correlations. The color represents the trait category.

## Discussion & Conclusions

- Our results identify the CNV burden sensitive functions impacting complex traits relevant to tissues (whole body)
- Deletions showed a higher proportion of FDR significant associations than duplications (24% vs 15%)
- CNVs disrupting **brain** gene sets, showed the **higher** associations with brain than non-brain traits (Del: 34% vs 17%, Dup: 17 % vs 11% )
- **CNV** burden correlations were higher than **common variant** genetic correlations across pairs of complex traits.
- Deletion and Duplication effect-size profiles show range of correlations (-0.69 to +0.33), implying that deletion and duplication have divergent phenotypic effects.
- Conclusion: We found that the majority of traits are dosage-sensitive. There is a convergent association between CNVs and common variants which proves that they have pleiotropic effects on the same pairs of traits with similar genetic-correlations.

## References

- Wiener et al, 2023
- Abdellaoui et al , 2021
- Bryois et al, 2020
- Kendall KM et al , 2016

- CNV Calling: https://github.com/MartineauJeanLouis/MIND-GENESPARALLELCNV
- Common Variant Genetic Correlations: https://www.nealelab.is/

## Acknowledgements

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